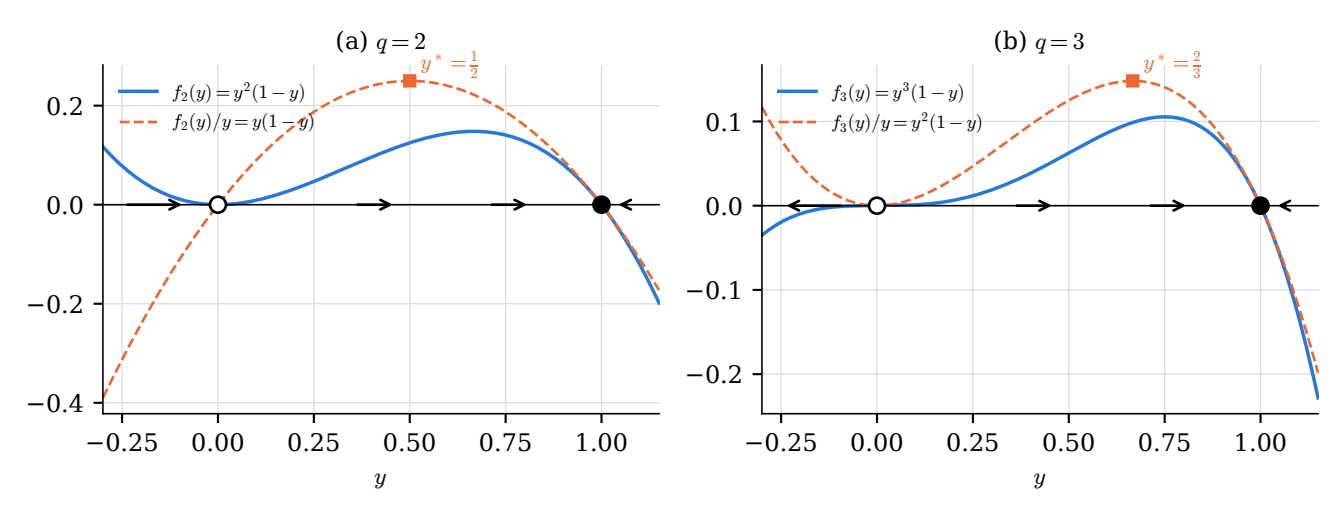
(a) $q = 2$
$y^* = \frac{1}{2}$
$f_2(y) = y^2(1-y)$
$f_2(y)/y = y(1-y)$
0.2
0.0
−0.2
−0.4
−0.25
0.00
0.25
0.50
0.75
1.00
y
(b) $q = 3$
$y^* = \frac{2}{3}$
$f_3(y) = y^3(1-y)$
$f_3(y)/y = y^2(1-y)$
0.1
0.0
−0.1
−0.2
−0.25
0.00
0.25
0.50
0.75
1.00
y